Set B *(Even Roll Numbers)*

Question 1: Directory Operations Create a

main directory using:

YourName_RollNo
Inside it perform:

a) Create subdirectories: Lab1, Lab2, Lab3
b) Move Lab3 inside Lab1
c) Create file studentinfo.txt inside Lab2
d) Copy studentinfo.txt into Lab1
e) Show directory tree structure

```
root@kali: /home/aman_singhal
Session  Actions  Edit  View  Help

┌──(root㉿kali)-[/home/aman_singhal]
└─# mkdir AmanSinghal_2301410004
```

```
root@kali: /home/aman_singhal/AmanSinghal_2301410004
Session  Actions  Edit  View  Help

┌──(root㉿kali)-[/home/aman_singhal]
└─# mkdir AmanSinghal_2301410004

┌──(root㉿kali)-[/home/aman_singhal]
└─# cd AmanSinghal_2301410004

┌──(root㉿kali)-[/home/aman_singhal/AmanSinghal_2301410004]
└─# mkdir Lab1, Lab2, Lab3

┌──(root㉿kali)-[/home/aman_singhal/AmanSinghal_2301410004]
└─# mv Lab3 Lab1/
```

```
┌──(root㉿kali)-[/home/aman_singhal/AmanSinghal_2301410004]
└─# touch Lab2,/studentinfo.txt

┌──(root㉿kali)-[/home/aman_singhal/AmanSinghal_2301410004]
└─# cp Lab2,/studentinfo.txt Lab1/

┌──(root㉿kali)-[/home/aman_singhal/AmanSinghal_2301410004]
└─# tree
.
├── Lab1
│   └── studentinfo.txt
├── Lab1,
├── Lab2,
│   └── studentinfo.txt
└── studentinfo.txt

4 directories, 3 files
```

```
┌──(root㉿kali)-[/home/aman_singhal/AmanSinghal_2301410004]
└─# ls -R
.:
Lab1  Lab1,  Lab2,  studentinfo.txt

./Lab1:
studentinfo.txt

./Lab1,:

./Lab2,:
studentinfo.txt
```

Question 2: Shell Script Writing Create a

shell script named:

scriptB_RollNo.sh Write

script to:

a) Accept two numbers from user
b) Display addition
c) Display subtraction
d) Display multiplication
e) Display greater number

Run the script successfully.

```
┌──(root㉿kali)-[/home/aman_singhal/AmanSinghal_2301410004]
└─# nano scriptB_2301410004.sh
```

```
root@kali: /home/aman_singhal/AmanSinghal_2301410004
Session  Actions  Edit  View  Help
  GNU nano 8.7.1                scriptB_2301410004.sh *
#!/bin/bash

echo "6"
read num1

echo "4"
read num2

echo "Addintion: $((num1 + num2))"
echo "Subtraction: $((num1 - num2))"
echo  "Multiplication: $((num1 * num2))"

if [$num1 -gt $num2]
then
   echo "$num2 is greater"
else
   echo "Both number are equal"
fi

^G Help        ^O Write Out   ^F Where Is    ^K Cut         ^T Execute
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify
```

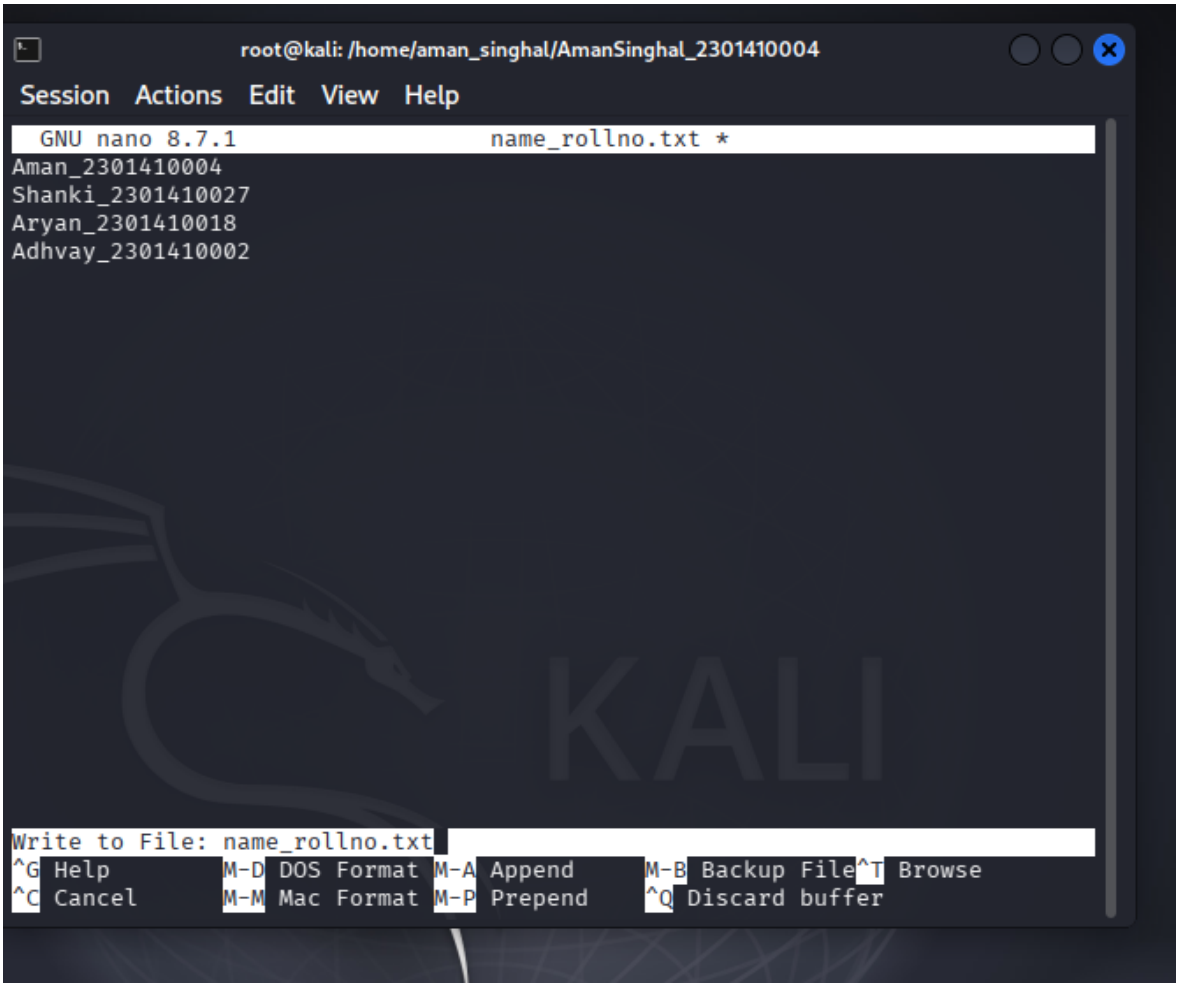
```
┌──(root㉿kali)-[/home/aman_singhal/AmanSinghal_2301410004]
└─# ./scriptB_2301410004.sh
6
12
4
2
Addintion: 14
Subtraction: 10
Multiplication: 24
```

Question 3: Search & Filters

Create file: names_RollNo.txt

Enter 5 names and perform:

a) Sort names alphabetically
b) Search names containing letter a
c) Count number of lines
d) Display first 3 lines
e) Display last 2 lines



```
┌──(root㉿kali)-[/home/aman_singhal/AmanSinghal_2301410004]
└─# nano name_rollno.txt

┌──(root㉿kali)-[/home/aman_singhal/AmanSinghal_2301410004]
└─# sort name_rollno.txt
Adhvay_2301410002
Aman_2301410004
Aryan_2301410018
Shanki_2301410027

┌──(root㉿kali)-[/home/aman_singhal/AmanSinghal_2301410004]
└─# grep "a" name_rollno.txt
Aman_2301410004
Shanki_2301410027
Aryan_2301410018
Adhvay_2301410002

┌──(root㉿kali)-[/home/aman_singhal/AmanSinghal_2301410004]
└─# grep -i "a" name_rollno.txt
Aman_2301410004
Shanki_2301410027
Aryan_2301410018
Adhvay_2301410002
```

```
┌──(root㉿kali)-[/home/aman_singhal/AmanSinghal_2301410004]
└─# wc -l name_rollno.txt
4 name_rollno.txt

┌──(root㉿kali)-[/home/aman_singhal/AmanSinghal_2301410004]
└─# head -n 3 name_rollno.txt
Aman_2301410004
Shanki_2301410027
Aryan_2301410018

┌──(root㉿kali)-[/home/aman_singhal/AmanSinghal_2301410004]
└─# tail -n 2 name_rollno.txt
Aryan_2301410018
Adhvay_2301410002
```